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EDUCATION

SAINT LOUIS UNIVERSITY Baguio City, Benguet
Bachelor of Science in Computer Science | Consistent Dean's Lister June 2024 - May 2027 (Expected)
Relevant Coursework: Data Structures & Algorithms, Software Engineering, Data Science, Artificial Intelligence, Web & App Development

PROJECTS

CORESET: VISION-ONLY EDGE AI BIOMECHANICAL REP-COUNTING MOBILE APP | *AI Model Code* | *App Source Code* Baguio City
Project Lead, Lead ML & Full Stack Engineer| *Python, React Native, PostgreSQL, PyTorch, MediaPipe, Tensorflow* Feb 2025 – May 2026

- Led the end-to-end development of an AI powered **mobile biomechanical analysis system** for real-time camera exercise recognition, repetition counting, skeletal overlay, and workout analytics across
- Engineered a **pose-based ML pipeline** using MediaPipe/BlazePose, 33 skeletal landmarks, 14 angle-based feature channels, 64-frame temporal windows, augmentation, denoising, and normalization across **1,000 balanced action sequences**.
- Developed and benchmarked BiLSTM and multi-task ST-GCN models, achieving **95.33% Top-1 accuracy**, 0.9935 Macro AUC, and 0.9527 Macro F1-score, then integrated the exported ST-GCN **TensorFlow Lite model** into a React Native/Expo app for **12.439 ms/sample mobile-ready inference**.

DATUBASE-PH: AI-DRIVEN MULTI-HORIZON POLITICAL DYNASTY FORECASTING | *Source Code* | Dashboard Baguio City
Project Lead, Lead AI Engineer & Researcher| *Python, PyTorch, Optuna, Scikit-Learn, Tableau* March 2025 – May 2026

- Led the development of an interpretable **Temporal Fusion Transformer (TFT)** forecasting system in **PyTorch** using a 23-year longitudinal dataset covering 81 Philippine provinces and **1,863 province-year observations**.
- Achieved a **2.98 MAE**, outperforming Naïve Persistence, SARIMA, and Ridge Regression baselines through advanced feature engineering and 50-trial Bayesian hyperparameter optimization with **Optuna**.
- Built an end-to-end AI analytics and interpretability pipeline using **Variable Selection Networks**, **Temporal Attention analysis,** and **Tableau** dashboards to forecast dynastic saturation and provincial institutional capture risk through 2031.

RXELLENCE: HEALTHCARE MANAGEMENT & PRESCRIPTION SYSTEM | *Source Code* Baguio City
Project Lead & Full-Stack Developer | *JavaScript, PHP, Node..js, HTML5, CSS3, Aiven MySQL, Chart.js, Docker* Oct 2025 – Dec 2025

- Delivered a healthcare management platform achieving **strict role-based access control** by architecting a hybrid **LAMP** and **Node.js** administrative layer.
- Streamlined doctor workflows, resulting in **highly efficient patient onboarding** and prescription staging, by engineering **dynamic CRUD operations** and **secure authentication pipelines**.
- Empowered medical administrators with actionable insights into patient demographics and medication trends by integrating **Chart.js** for advanced, **interactive data visualization**.

NAVI-NOW: SMART SHUTTLE RESERVATION SYSTEM | *Source Code* Baguio City
Project Lead & Full-Stack Developer | *JavaScript, Node.js, Express.js, HTML, Typescript, Aiven MySQL* Sept 2025 – Dec 2025

- Modernized traditional commuter travel, resulting in **frictionless boarding experiences**, by developing a **real-time shuttle reservation platform** with **live capacity tracking**.
- Eliminated physical ticketing bottlenecks and secured transactions by implementing an **automated QR-code generation system** and an **integrated payment gateway**.
- Ensured highly scalable application architecture and clear technical direction for the development team by **engineering end-to-end system workflows** and authoring comprehensive functional requirements.

COMMUNITY: A VOLUNTEER SERVICE TRACKER | *Source Code* Baguio City
Project Lead & Full-Stack Developer | *Java, JavaFX, FXML, Aiven Aiven MySQL* June 2025 – May 2025

- Architected a scalable, cloud-hosted **SQL database (Aiven SQL)**, designing complex **Entity-Relationship Diagrams** (ERDs) to optimize data queries and manage highly relational, many-to-many dependencies across users, groups, and concurrent events.
- Engineered a full-stack **Java** and **JavaFX** application using **MVC architecture**, implementing strict Role-Based Access Control (RBAC) pipelines to securely process dynamic data for distinct volunteer and administrator modules.
- Directed the end-to-end **Agile development** lifecycle via **GitLab**, serving as the primary technical lead to conduct rigorous code reviews, debug cross-functional pull requests, and enforce strict version control for a highly stable deployment.

WHAT'S THE WORD: NETWORKED MULTIPLAYER WORD GAME | *Source Code* Baguio City
Full Stack Developer & Software Engineer | *Java, JavaFX, FXML, Aiven MySQL, CORBA* March 2025 – June 2025

- Engineered a high-throughput, **multithreaded client-server architecture** utilizing Java Sockets, ensuring thread-safe data handling and zero-desync state synchronization for multiple concurrent game lobbies.
- Architected a **Role-Based Access Control (RBAC) backend** with complete end-to-end CRUD pipelines, empowering administrators to securely manage player lifecycles and persistently store user states in a normalized MySQL database.
- Built a **dynamic configuration engine** that parses local file I/O to inject live game rules (e.g., matchmaking timeouts, round limits) directly into the active server state, allowing for real-time pacing adjustments without requiring deployment downtime.

SKILLS AND INTERESTS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS, PHP, C, C++
Frameworks & Tools: React Native, NodeJS, ExpressJS, MySQL, PostgreSQL, PyTorch, TensorFlow, Docker, Git, Tableau, Figma, VSCode
Interests: Software Engineering, Full-Stack Development, Machine Learning, Deep Learning, System Design, Data Analysis, UI/UX